

§12 Principal Angles between Per-Sample Curvature Subspaces

How the min / max / mean principal angles are computed in code

eos_widget — _sec12_payload

1 What is being measured

For each training sample i the per-sample function Hessian $Q_i = \nabla^2 f_i(\theta) \in \mathbb{R}^{p \times p}$ (the Hessian of the ground-truth-label network output f_i with respect to the p parameters) has an eigendecomposition. We summarise each Q_i by a low-dimensional *curvature subspace*: the span of its top- k eigenvectors (largest eigenvalues) together with its bottom- k eigenvectors (most negative eigenvalues). The diagnostic then asks: *how aligned are these curvature subspaces across different samples?* The alignment of two subspaces is quantified by their **principal angles**, and Panels 1–3 plot the $N \times N$ grid of pairwise angles together with its min, max, and mean over sample pairs, as a function of training step.

2 The code

The computation is the function `pa(k)` together with `offstats`, reproduced here (ASCII-transliterated; TV,BV are the per-sample top/bottom eigenvectors of shape (N, K, p)):

```
offm = ~torch.eye(N, dtype=torch.bool, device=dev) # off-diagonal (i != j) mask

# principal angles (gauge-free: svdvals invariant to eigenvector sign) for k=1,2,5,10,kfull
def pa(k):
    kk = max(1, min(k, K))
    E = torch.cat([TV[:, :kk, :], BV[:, :kk, :]], dim=1) # (N, 2kk, p)
    sv = torch.linalg.svdvals(torch.einsum('iap,jbp->ijab', E, E)).clamp(-1.0, 1.0)
    return (torch.acos(sv) * (180.0 / pi)).mean(dim=2) # (N, N) mean angle per pair

def offstats(g): # min/max/mean over off-diagonal pairs (i != j)
    s = g[offm]
    return float(s.min()), float(s.max()), float(s.mean())
```

We now translate each line into mathematics.

3 Step 1: the per-sample subspace

For a chosen rank k let $kk = \min(k, K)$. Stack sample i 's top- kk and bottom- kk eigenvectors as the rows of

$$E_i^{(k)} = \begin{bmatrix} u_{i,1}^\top \\ \vdots \\ u_{i,kk}^\top \\ u_{i,-1}^\top \\ \vdots \\ u_{i,-kk}^\top \end{bmatrix} \in \mathbb{R}^{2kk \times p}, \quad \mathbf{E} = \text{cat}([\text{TV}[:, :kk], \text{BV}[:, :kk]]). \quad (1)$$

Each u is a unit vector in parameter space $\mathbb{R}^p$, and within a sample the $2kk$ eigenvectors are mutually orthonormal (they are Ritz vectors of one fully-reorthogonalised Lanczos basis), so $E_i^{(k)}$ has orthonormal rows. It is therefore an orthonormal basis for the $2kk$ -dimensional subspace

$$\mathcal{S}_i^{(k)} = \text{span}\{u_{i,1}, \dots, u_{i,kk}, u_{i,-1}, \dots, u_{i,-kk}\} \subseteq \mathbb{R}^p, \quad \dim \mathcal{S}_i^{(k)} = 2kk. \quad (2)$$

In the code, k ranges over $\{1, 2, 5, 10, k_{\text{full}}\}$ with $k_{\text{full}} = \min(15, K)$.

4 Step 2: principal angles between two samples

For a pair (i, j) , the einsum `'i ap, j bp -> i j ab'` forms the cross-Gram matrix

$$M_{ij} = E_i^{(k)} (E_j^{(k)})^\top \in \mathbb{R}^{2kk \times 2kk}, \quad (M_{ij})_{ab} = \langle e_{i,a}, e_{j,b} \rangle, \quad (3)$$

i.e. all pairwise inner products between the eigenvectors of sample i and those of sample j . Let its singular values be

$$\sigma_1 \geq \sigma_2 \geq \dots \geq \sigma_{2kk}, \quad \sigma_\ell = \sigma_\ell(M_{ij}) \in [0, 1] \quad (\text{svdvals, clamped to } [-1, 1]). \quad (4)$$

By the variational (CS / Jordan) characterisation of **principal angles** $0 \leq \theta_1 \leq \dots \leq \theta_{2kk} \leq \frac{\pi}{2}$ between two subspaces with orthonormal bases U, V , the singular values of $U^\top V$ are exactly the cosines of those angles:

$$\cos \theta_\ell = \sigma_\ell, \quad \implies \quad \theta_{ij,\ell}^{(k)} = \arccos(\sigma_\ell) \cdot \frac{180}{\pi} \text{ (degrees)}. \quad (5)$$

The clamp to $[-1, 1]$ only guards against a singular value numerically exceeding 1, which would make $\arccos$ undefined. Recall the recursive meaning: $\cos \theta_1$ is the largest possible correlation $\langle u, v \rangle$ over unit vectors $u \in \mathcal{S}_i$, $v \in \mathcal{S}_j$; each subsequent angle maximises correlation subject to orthogonality to the previous principal directions.

Gauge invariance. Eigenvectors are only defined up to sign (and up to an orthogonal rotation inside any degenerate eigenspace). Replacing $E_i \mapsto O_i E_i$ for orthogonal O_i leaves the singular values of $M_{ij} = E_i E_j^\top \mapsto O_i E_i E_j^\top O_j^\top$ unchanged. Hence $\theta_{ij,\ell}^{(k)}$ depends only on the *subspaces* $\mathcal{S}_i^{(k)}, \mathcal{S}_j^{(k)}$, not on the eigenvector signs or basis choice — this is the “gauge-free” comment, and is why we use singular values of the cross-Gram rather than raw dot products.

5 Step 3: two levels of aggregation

Inner mean (`.mean(dim=2)`). The $2kk$ angles of a single pair are averaged into one scalar, giving the $N \times N$ grid

$$\bar{G}_{ij}^{(k)} = \frac{1}{2kk} \sum_{\ell=1}^{2kk} \theta_{ij,\ell}^{(k)} \in [0^\circ, 90^\circ]. \quad (6)$$

This $\bar{G}^{(k)} \in \mathbb{R}^{N \times N}$ is the **Panel-1 heatmap**; by construction $\bar{G}_{ii}^{(k)} = 0$ (a subspace makes a zero angle with itself).

Outer min / max / mean (offstats). The mask `offm = {(i, j) : i ≠ j}` removes the trivial diagonal, and the three reported statistics are taken over the $N(N-1)$ off-diagonal ordered pairs:

$$\theta_{\min}^{(k)} = \min_{i \neq j} \bar{G}_{ij}^{(k)}, \quad \theta_{\max}^{(k)} = \max_{i \neq j} \bar{G}_{ij}^{(k)}, \quad \theta_{\text{mean}}^{(k)} = \frac{1}{N(N-1)} \sum_{i \neq j} \bar{G}_{ij}^{(k)}. \quad (7)$$

These are the fields `mn`, `mx`, `me`; Panels 2–3 plot them versus the training step (one curve per rank k).

Whole-grid mean (gm). The Panel-1 title uses the mean over *all* N^2 entries, including the N diagonal zeros, so it is just the off-diagonal mean diluted by the diagonal:

$$\text{gm}^{(k)} = \frac{1}{N^2} \sum_{i,j} \bar{G}_{ij}^{(k)} = \frac{N(N-1)}{N^2} \theta_{\text{mean}}^{(k)} = \frac{N-1}{N} \theta_{\text{mean}}^{(k)}. \quad (8)$$

6 Interpretation

All quantities are in degrees: 0° means two samples share the same curvature subspace, 90° means the subspaces are orthogonal.

- $\theta_{\min}^{(k)}$ — the most-aligned pair of samples (smallest angle): the strongest shared top/bottom curvature directions in the dataset.
- $\theta_{\max}^{(k)}$ — the most-misaligned pair.
- $\theta_{\text{mean}}^{(k)}$ — the typical pairwise misalignment across all sample pairs.

Tracking these versus the training step shows whether the per-sample curvature subspaces become more or less aligned as progressive sharpening / edge-of-stability proceeds.